Rethinking case attraction on Ancient Greek infinitive clauses

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1 Introduction

Case attraction on infinitive clauses (contrast (1-a) with (1-b) below) has been analysed over the recent years as a idiosyncratic and strictly syntactic construction (Tantalou 2003, Spyropoulos 2005 and Sevdali 2013a,b), lacking many parallels across either ancient or modern languages and across other linguistic *phenomena*.

(1) a. συμβουλεύει τῷ Ξενοφῶντι ἐλθόντα εἰς Δελφοὺς ἀνακοινῶσαι
advice.3SG X.DAT.SG going.ACC.SG to-Delphi ask.INF
τῷ θεῷ περὶ τῆς πορείας.
the-god.DAT.SG about-the-travel
He advices Xenophon to go to Delphi and ask the god about the travel. (Xen. Anab. 3.1.5)

b. ἀφῆκε μοι ἐλθόντι πρὸς ὑμᾶς λέγειν
allowed.3SG PRON.1SG.DAT going.DAT.SG in-front-of-you say.INF
τἀληθῆ.
the-truth.ACC.
He allowed me to go and speak the truth in front of you. (Xen. Hell. 6.1.13)

The evidence I have brought forward in previous works (Geraldes 2020, 2021, 2025) shows that, at least in terms of distribution, case attraction does seems to covary with factors associated with distinct semantic/pragmatic features, namely word-order, and semantic and syntactic class of the matrix and infinitival verb.

The evidence across natural languages shows that *non-canonical* agreement / concord takes place in a non-deterministic fashion as semantic or pragmatic features of the sentence appear more marked, which is similar to the contexts associated with case attraction assumed by grammarians as early as Buttmann (1826), contexts in which some sort of *emphasis* is assigned to the target of attraction makes it more likely to be attracted. Although the intuition seems to be sound, there is little to

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no specificity in what is denoted by *emphasis* and the explanation is prone to *ad hoc* interpretations.

2 A stoachastic approach

Using data from literary sources of Classical Greek, including oratory speeches, drama, historiography and philosophical dialogues from Attic and Jonic sources, I provide a data driven quantitative analysis of the contexts in which case attraction is a possible agreement / concord resolution. The addition of Jonic sources is due to the fact that it has been assumed that case attraction is more common if not the rule in the Attic dialect (e.g. Buttmann 1826, *passim.* and Cooper and Krüger 1997, *ad loc.*). The data has been collected and annotated in a combination of manual and computational methods using the Diorisis Ancient Greek Corpus (Vatri and McGillivray 2018).

A quantitative analysis of case attraction requires caution in the methodological approach, as semantic and pragmatic features are often latent, i.e. not explicitly present in the morpho-phonological level, and the use of proxies such as word classes, constituent distance and word order may hinder the quality of the results and the causal inference built upon them. As such, our analysis will rely on the causality analysis (e.g. Pearl 2009) and Bayesian modeling (e.g. McElreath 2020), which I argue could enhance the results of data driven linguistic research by including at the quantitative analysis the qualitative knowledge built on the Ancient Greek and general linguistics. The analysis must be thus twofold. Firstly, it assess how the linguistic and extralinguistic factors could reasonably be causally linked with case attraction, so as to inform the statistical modeling and tell what effects are possible to estimate from the data. Later, a general linear model is build as to adequately estimate the direct effects of semantic and pragmatic factors on case attraction and the interactions between linguistic and extralinguistic variables.

3 A causal model for case attraction

- Controllers:
  - C-1: <u>controller present</u> > controller absent
  - C-2: <u>controller has overt expression of agreement features</u> > controller has covert expression of agreement features
    - “A canonical controller marks at least as many distinctions as the target”
  - C-3: consistent controller > <u>hybrid controller</u>
  - C-4: <u>controller’s part of speech is irrelevant</u> > is relevant (given the domain)
- Targets:
  - C-5: <u>inflectional marking (affix)</u> > clitic > free word
  - C-6: obligatory > <u>optional</u>

    - C-7: regular > suppletive
    - C-8: alliterative > opaque
    - C-9: productive marking of agreement > sporadic marking
    - C-10: target always agree > target agrees only when controller is absent
    - C-11: target agrees with a single controller > agrees with more than one controller
    - C-12: target has no choice of controller > target has choice of controller (is 'trigger-happy')
    - C-13: target's part of speech is irrelevant > is relevant (given the domain)
- Domains:
    - C-14: asymmetric > symmetric
    - C-15: local domain > non-local domain
    - C-16: domain is one of a set > single domain
- Features:
    - C-17: feature is lexical > non-lexical
    - C-18: features have matching values > non-matching values
    - C-19: no choice of feature value > choice of value
- Conditions:
    - C-20: no conditions > conditions

By assuming that case attraction (A) is a sort of *non-canonical agreement/concord*, it must be expected that factors known cross-linguistically to trigger these *phenomena* also affect A. Drawing a list of conditioning factors for *non-canonical agreement* from Corbett (2006, pp. 176–205), we should thus expect that the following factors would change the likelihood of A. 1. thematic role (Θ) and animacy (An); 2. grammatical relation (GR) and case (K); and 3. communicative function (CF), and precedence ($Prec$).

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